

B. Configurational averages

As formulated the TLS potential corresponds to a particular realization of a distribution of TLS impurities, which are located at random positions in the otherwise crystalline superconductor. The orientation of the two-sites of double-well potential is also a random variable. Finally, the strain introduced by the random distribution of impurities induces a stochastic distribution of tunnel barriers, distances between local minima, asymmetries in the local minima, and thus a statistical distribution of tunnel splittings of the TLS impurities. This configurational randomness is in addition to the statistical distribution of states defined by the external thermal environment in contact with the superconductor. Thus, we consider the interaction of electrons and pairs of electrons with a dilute random distribution of TLS impurities following methods introduced by Edwards [32], Anderson [11] and Abrikosov and Gorkov [12]. In particular, we treat the positions of the TLS impurities, $\{\mathbf{R}_j | j \in 1, 2, \dots, N\}$, as random variables described by an uncorrelated joint probability distribution, $P(\{\mathbf{R}_j\})$. This allows us to carry out configurational averages of the perturbation expansion of the Gorkov Green's function in powers of $H_{\text{e-TLS}}$. The resulting configurational average results in coarse-grained translational invariance for configurational averages of all terms of the perturbation expansion for the Green's function. Thus,

$$\overline{\hat{G}_{k,k'}(\{\mathbf{R}_j\})} = \prod_{j=1}^{N_{\text{TLS}}} \int \frac{d^3 R_j}{V} \hat{G}_{k,k'}(\{\mathbf{R}_j\}) = \delta_{\mathbf{k},\mathbf{k}'} \hat{G}_k. \quad (23)$$

We generalize the conventional theory of dilute random impurities [11, 12, 32] to include configurational averaging over TLS impurities, including the orientation of the tunneling path, as well as the distribution of tunnel splittings. Averaging over positions of the TLS impurities follows the Edwards, Anderson and Abrikosov-Gorkov theory for a homogeneous distribution of uncorrelated randomly located impurities. The vertex for quasiparticle-impurity scattering depends on the random positions of the trapping potential of the TLS impurities, $\{R_j\}$, i.e. $A_{\mathbf{k}',\mathbf{k}}^{ab}(j) = e^{i(\mathbf{k}'-\mathbf{k})\cdot\mathbf{R}_j} \hat{A}_{\mathbf{k}',\mathbf{k}}^{ab}(j)$. Thus, after summing over random phase factors, $e^{i(\mathbf{k}'-\mathbf{k})\cdot\mathbf{R}_j}$, translational invariance is recovered for the configurational averaged Green's function. After configurational averaging over the random positions of TLS impurities the scattering amplitude still depends on the *orientation* of the tunneling path, and thus the orientation of the tunneling path relative to the momentum $\mathbf{k}$. Here we assume that the orientation of tunneling path of the TLS impurities is randomly distributed with a uniform probability distribution. As a result coarse-grained rotational invariance is recovered for the configurational averaged propagator.

Configurational averaging the 1st and 2nd order terms in the perturbation expansion of the Nambu Green's function shown in Fig. 3 generates the set of Feynman diagrams shown in Fig. 4. These processes separate into two classes: those in which a quasiparticle exchanges energy with a TLS impurity, and those that do not. Diagrams Fig. 4(a) and Fig. 4(b) are processes with no transfer of momentum or exchange of energy between the TLS impurities and quasiparticles. These terms simply renormalize the chemical potential. Diagram Fig. 4(c) is the 2nd order Born scattering process from the static contribution to the TLS impurity potential. Diagrams Fig. 4(d),(e) and (f), in which all pseudo-Fermion lines form loops connected at a single vertex, are quasiparticle self-energy terms that depend on the occupations of the TLS levels, but do not involve exchange of energy between quasiparticles and the TLS impurities. The term shown in Fig. 4(g) is the two-loop pseudo-Fermion correction to quasiparticle and Cooper pair self energy. In this process the pseudo-Fermion lines connect initial and final vertices, and thus describe transitions between the excited and ground states of the TLS. As a result energy is transferred between the TLS impurities and quasiparticles. This term plays the central role in pair-breaking and pair enhancement via quasiparticle-TLS scattering.

IV. SELF ENERGIES

The corrections to the Gorkov propagator for quasiparticles and pairs resulting from interactions between electrons and TLS impurities are encoded in the self energy functional via the Dyson equation,

$$\hat{G}_k^{-1} = \hat{G}_k^{(0)-1} - \hat{\Sigma}_k = i\tilde{\varepsilon}_n \hat{1} - \xi_{\mathbf{k}} \hat{\tau}_3 - \tilde{\Delta}(is_y)(-i\hat{\tau}_2), \quad (24)$$

where $\hat{G}_k \equiv \hat{G}(\mathbf{k}, \varepsilon_n)$ and the self-energy can be expressed as $\hat{\Sigma}_k = \Sigma_\varepsilon \hat{1} + \Sigma_\Delta(is_y)(-i\hat{\tau}_2)$ in Nambu space. The renormalized Matsubara energy and superconducting order parameter are

$$i\tilde{\varepsilon}_n = i\varepsilon_n - \Sigma_\varepsilon, \quad (25)$$

$$\tilde{\Delta} = \Delta + \Sigma_\Delta. \quad (26)$$

The functional form of the contributions to the self energy functional follow from the Feynman diagrams.

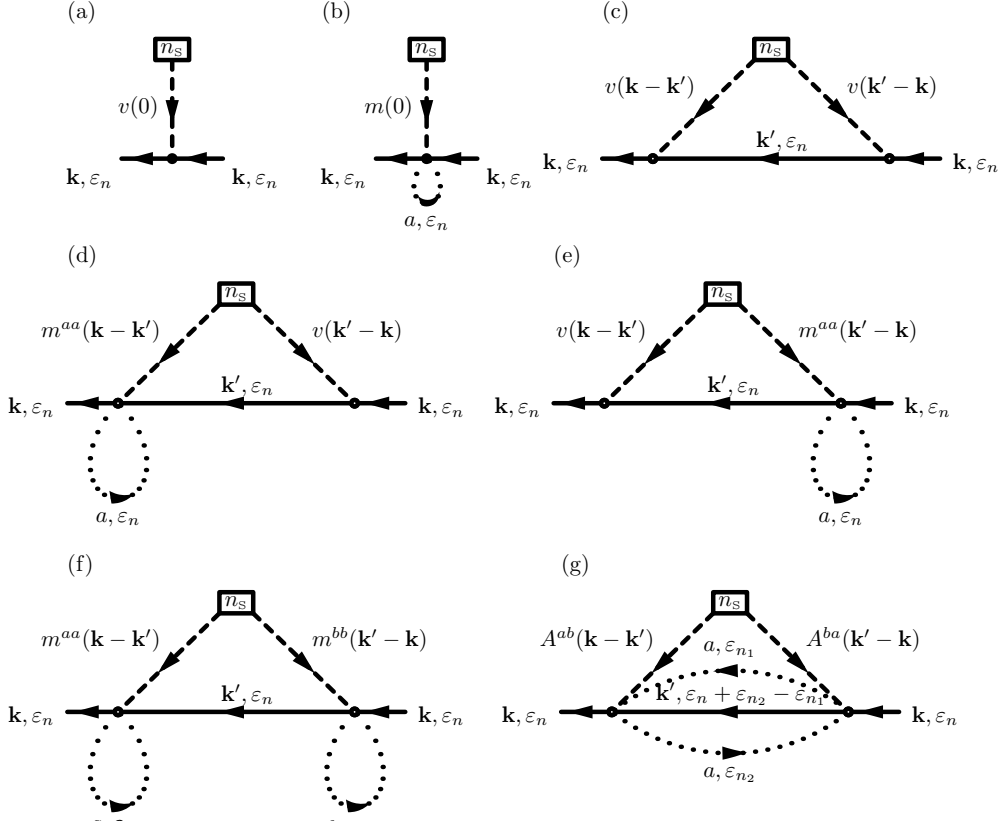


FIG. 4. All contributions are proportional to the mean TLS density, $n_S = N_S/V$. The top row are the 1^{st} and 2^{nd} order self energy diagrams from the static contribution to the TLS impurity scattering matrix element. The second row are one-loop pseudo-Fermion corrections to the quasiparticle-TLS self-energy. These terms depend on both the static and dynamical interactions of the TLS impurities. The third row are two-loop pseudo-Fermion corrections to the quasiparticle-TLS self-energy. In particular, the last diagram describes multiple scattering by a TLS impurity including transitions between the excited and ground states of the TLS impurity.

A. Elastic scattering by TLS impurities

For the diagrams that do not involve energy exchange between quasiparticles and TLS impurities the terms can be expressed in terms of the various matrix elements, v , m , and n , and the pseudo-Fermion loops attached to the corresponding vertices. For a pseudo-Fermion loop terms containing the n matrix elements vanish because σ_x is off-diagonal. Thus, the vertex pairs are (vv) , (vm) , (mv) and (mm) , as shown in Figs. 4(c),(d),(e) and (f). The matrix elements, $v_{\mathbf{k},\mathbf{k}'}$, $m_{\mathbf{k},\mathbf{k}'}$, $n_{\mathbf{k},\mathbf{k}'}$, are slowly varying functions of the magnitude of the momenta, whereas the spectral weight of the propagator is concentrated near the Fermi level. Thus we can evaluate the matrix elements at $|\mathbf{k}| = k_f$ and $|\mathbf{k}'| = k_f$, in which case are functions of the directions, $\hat{\mathbf{k}}$ and $\hat{\mathbf{k}'}$, on the Fermi surface. Similarly, the normal-state density of states can be evaluated at the Fermi level, $N(\xi_{\mathbf{k}}) \rightarrow N(0)$. The corresponding self-energies reduce to,

$$\hat{\Sigma}_k^{(vv)} = \frac{1}{V} \sum_j N(0) \langle v_{\mathbf{k},\mathbf{k}'} v_{\mathbf{k}',\mathbf{k}} \rangle_{\Omega} \int d\xi_{\mathbf{k}'} \hat{\tau}_3 \hat{G}(\mathbf{k}', i\varepsilon_n) \hat{\tau}_3, \quad (27)$$

$$\hat{\Sigma}_k^{(vm)} = -2 \frac{1}{V} \sum_j N(0) \sum_a \langle v_{\mathbf{k},\mathbf{k}'} m_{\mathbf{k}',\mathbf{k}}^{aa} \rangle_{\Omega} T \sum_{n_1} D_{j,a}(\varepsilon_{n_1}) \int d\xi_{\mathbf{k}'} \hat{\tau}_3 \hat{G}(\mathbf{k}', i\varepsilon_n) \hat{\tau}_3, \quad (28)$$

$$\hat{\Sigma}_k^{(mm)} = \frac{1}{V} \sum_j N(0) \sum_{ab} \langle m_{\mathbf{k},\mathbf{k}'}^{aa} m_{\mathbf{k}',\mathbf{k}}^{bb} \rangle_{\Omega} T \sum_{n_1} D_{j,a}(\varepsilon_{n_1}) T \sum_{n_2} D_{j,b}(\varepsilon_{n_2}) \int d\xi_{\mathbf{k}'} \hat{\tau}_3 \hat{G}(\mathbf{k}', i\varepsilon_n) \hat{\tau}_3, \quad (29)$$

where $k \equiv (\mathbf{k}, i\varepsilon_n)$, and $\langle \dots \rangle_{\Omega} = \int \frac{d\Omega_{\hat{\mathbf{k}}}}{4\pi} \int \frac{d\Omega_{\hat{\mathbf{k}'}}}{4\pi} (\dots)$ averages $\mathbf{k}$ and $\mathbf{k}'$ over the Fermi surface. The sign in Eq. (28) comes from the single pseudo-Fermion loop and the factor of 2 is that diagrams (d) and (e) in Fig. 4 are equivalent after angular integration.

The pseudo-Fermion loops are evaluated using Cauchy's theorem to express the Matsubara sum as a contour integral of the Fermi function and the pseudo-Fermion propagator, analytically continued to the complex energy plane, enclosing the poles, $z_n = i\varepsilon_{n_1}$ of the Fermi function, $n_F(z) = 1/(e^{\beta z} + 1)$, then deforming the contour to encircle the pole of $D_{j,a}(z)$ at $z = \varepsilon_{j,a} - \mu_f$. The resulting residue gives the sum,

$$T \sum_{n_1} D_{j,a}(\varepsilon_{n_1}) = n_F(\varepsilon_{j,a} - \mu_f) = \frac{1}{1 + i e^{\beta \varepsilon_{j,a}}} = \frac{e^{-\beta \varepsilon_{j,a}} - i}{e^{\beta \varepsilon_j} + e^{-\beta \varepsilon_j}}. \quad (30)$$

The denominator is the partition function for the two levels of the TLS $\varepsilon_{j,g} = -\varepsilon_j$ and $\varepsilon_{j,e} = \varepsilon_j > 0$, i.e. $Z_j = e^{\beta \varepsilon_j} + e^{-\beta \varepsilon_j}$, and thus, $N_{j,a} = e^{-\beta \varepsilon_{j,a}}/Z_j$ is the equilibrium occupation of the two states of the j^{th} TLS. Thus,

$$T \sum_{n_1} D_{j,a}(\varepsilon_{n_1}) = N_a(\varepsilon_j) - i/Z(\varepsilon_j). \quad (31)$$

Using $m_{\mathbf{k}',\mathbf{k}}^{aa} = m_{\mathbf{k}',\mathbf{k}} \sigma_z^{aa}$ and carrying out the sum over level indices in Eqs. (28) and (29), we obtain,

$$\sum_a \sigma_z^{aa} T \sum_{n_1} D_{j,a}(\varepsilon_{n_1}) = N_e(\varepsilon_j) - N_g(\varepsilon_j). \quad (32)$$

Thus, the scattering terms in Eqs. (28) and (29) depend on the level populations of the states of the TLS, but these scattering processes do not change the level population of the TLS, and thus involve only elastic scattering of quasiparticles off the TLS impurities. This is evident in that the Matsubara energy of the intermediate propagator in all of the above processes is $i\varepsilon_n$, i.e. for these scattering processes there is no energy exchange with the TLS impurity at either vertex.

1. Quasiclassical Propagator and Self Energy

The Nambu matrix propagator appearing in Eqs. (27)-(29) is integrated over the magnitude of the momentum $|\mathbf{k}|$ near the Fermi surface, or equivalently the normal-state quasiparticle excitation energy, $\xi_{\mathbf{k}} = v_f(|\mathbf{k}| - k_f)$,

$$\hat{\mathcal{G}}(\hat{\mathbf{k}}, \varepsilon_n) \equiv \int d\xi_{\mathbf{k}} \hat{\tau}_3 \hat{G}(\mathbf{k}, i\varepsilon_n). \quad (33)$$

This is Eilenberger's quasiclassical propagator which is reduced to a function of position on the Fermi surface, $\hat{\mathbf{k}}$. Note also the factor of $\hat{\tau}_3$ as it is important in Eilenberger's reduction of the Dyson equation to a transport-like equation for the matrix $\hat{\mathcal{G}}(\hat{\mathbf{k}}, \varepsilon_n; \mathbf{r})$ defined along trajectories specified by the normal-state group velocity, $\mathbf{v}_{\mathbf{k}} = v_f \hat{\mathbf{k}}$, and locally for inhomogeneous equilibrium states,

$$\left[i\varepsilon_n \hat{\tau}_3 - \hat{\Delta}(\hat{\mathbf{k}}; \mathbf{r}) - \hat{\Sigma}(\hat{\mathbf{k}}, \varepsilon_n; \mathbf{r}), \hat{\mathcal{G}}(\hat{\mathbf{k}}, \varepsilon_n; \mathbf{r}) \right] + i\mathbf{v}_{\mathbf{k}} \cdot \nabla_{\mathbf{r}} \hat{\mathcal{G}}(\hat{\mathbf{k}}, \varepsilon_n; \mathbf{r}) = 0, \quad (34)$$

where $\hat{\Delta}(\hat{\mathbf{k}}; \mathbf{r}) = \Delta(\hat{\mathbf{k}}; \mathbf{r}) i s_y \hat{\tau}_1$ is the mean-field order parameter, which is defined here by the weak-coupling gap equation,

$$\Delta(\hat{\mathbf{k}}; \mathbf{r}) = \int \frac{d\Omega_{\mathbf{k}'}}{4\pi} g(\hat{\mathbf{k}}, \hat{\mathbf{k}}') T \sum_{\varepsilon_n}^{\omega_c} \mathfrak{F}(\hat{\mathbf{k}}', \varepsilon_n; \mathbf{r}), \quad (35)$$

where $g(\hat{\mathbf{k}}, \hat{\mathbf{k}}')$ is the pairing interaction within the bandwidth of attraction, $|\varepsilon_n| \leq \omega_c$, and $\mathfrak{F}(\hat{\mathbf{k}}, \varepsilon_n; \mathbf{r})$ is Gorkov's anomalous propagator in the quasiclassical limit, i.e. the off-diagonal component of the matrix propagator, $\hat{\mathcal{G}}(\hat{\mathbf{k}}, \varepsilon_n; \mathbf{r})$. The quasiclassical self energy is evaluated for momentum on the Fermi surface, and for the configurationally averaged TLS self energy defined by the diagrams in Fig. 4, is post-multiplied by $\hat{\tau}_3$, i.e. $\hat{\Sigma}(\hat{\mathbf{k}}, \varepsilon_n) \equiv \hat{\Sigma}_k \hat{\tau}_3$. The Nambu matrix structure of the TLS self energies is determined by the Nambu matrix structure of the propagator, i.e. $\hat{\mathcal{G}} = \mathcal{G} \hat{\tau}_3 + \mathfrak{F} i s_y \hat{\tau}_1$. Thus, we can express $\hat{\Sigma} = \Sigma_\varepsilon \hat{\tau}_3 + \Sigma_\Delta i s_y \hat{\tau}_1$. The normalization of the propagator, which is absent in Eq. (34), is enforced by the Eilenberger's normalization condition [33],

$$\left[\hat{\mathcal{G}}(\hat{\mathbf{k}}, \varepsilon_n) \right]^2 = -\pi^2 \hat{1}. \quad (36)$$

In what follows we consider isotropic ("s-wave") pairing of conduction electrons near an isotropic Fermi surface, i.e. $g(\hat{\mathbf{k}}, \hat{\mathbf{k}}') \rightarrow g$ and $\mathbf{v}_{\mathbf{k}} = v_f \hat{\mathbf{k}}$. Thus the mean field order parameter, propagator and self energy are all isotropic on the Fermi surface. We also consider only homogeneous superconducting states in zero magnetic field, in which case the propagator and self energy are spatially uniform.